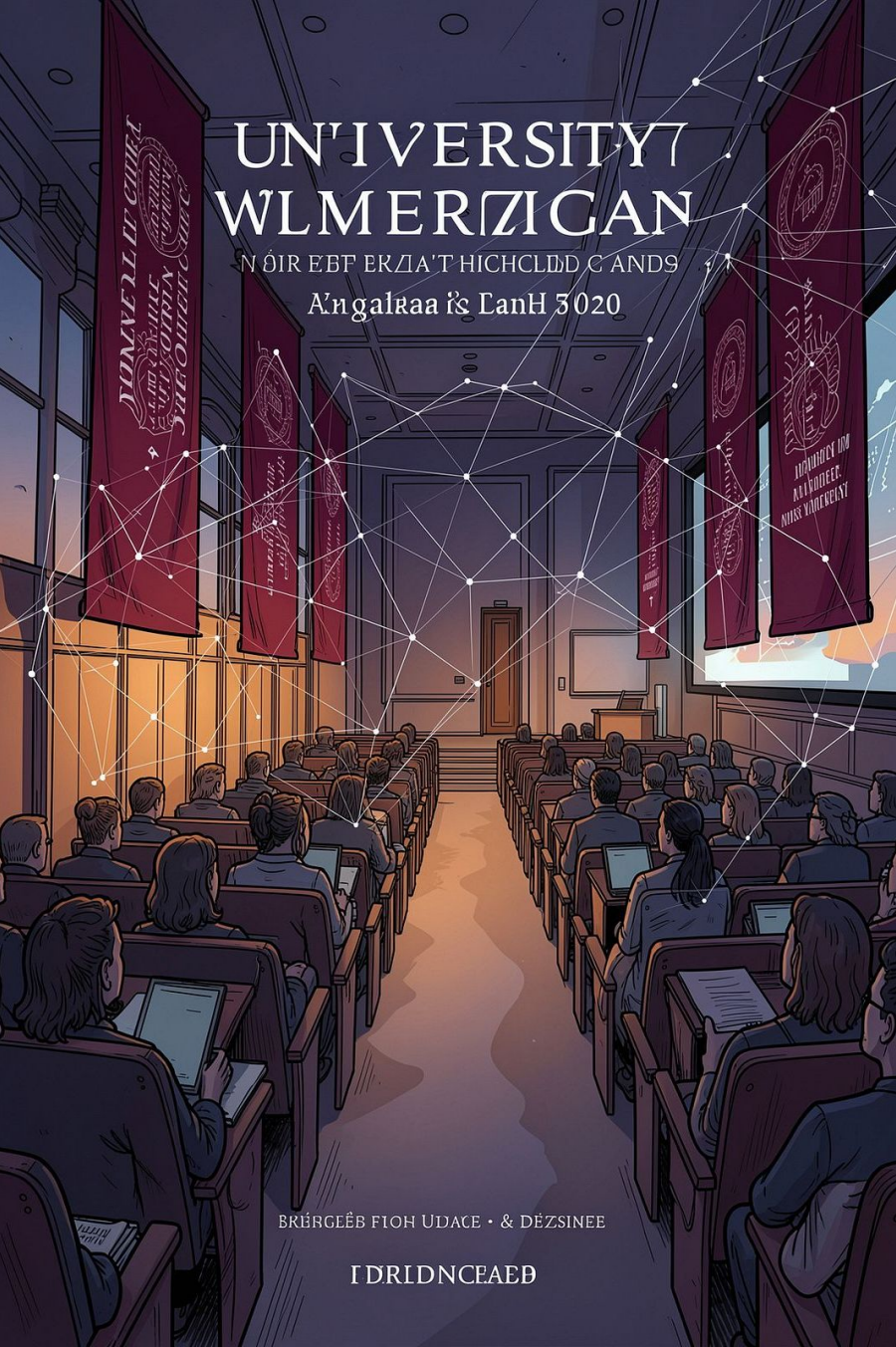




UNIVERSITY WOLLAM COLLEGE

WOLLAM COLLEGE
WOLLAM, VIRGINIA
AUGUST 15, 2020

Design & Implementation of a DBMS for Research Project, Publication & Grant Management System

“A centralized database system for managing research projects, publications, researchers and grants.”

Database Management System - Review 1

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BRIDGE FLOW UJACE • & DESIGNEE

INDIAN

Made with GAMMA

Problem background

Universities manage multiple research projects, publications and grants using separate spreadsheets, documents and manual records. This causes duplicated records, missed deadlines, inconsistent metadata, and poor reporting for audits and funding bodies.



Time lost

Manual updates and chasing collaborators reduce researcher productivity.



Data risk

Inconsistent metadata, version drift and missing provenance undermine reproducibility.



Poor analytics

No single source of truth for reporting research outputs and fund utilisation.

Why manual management fails

Scalability

Spreadsheets and email do not scale to departmental volume and cross-institution collaborations.

Traceability

No reliable provenance for submissions, approvals or budget changes — hard to audit or reproduce decisions.

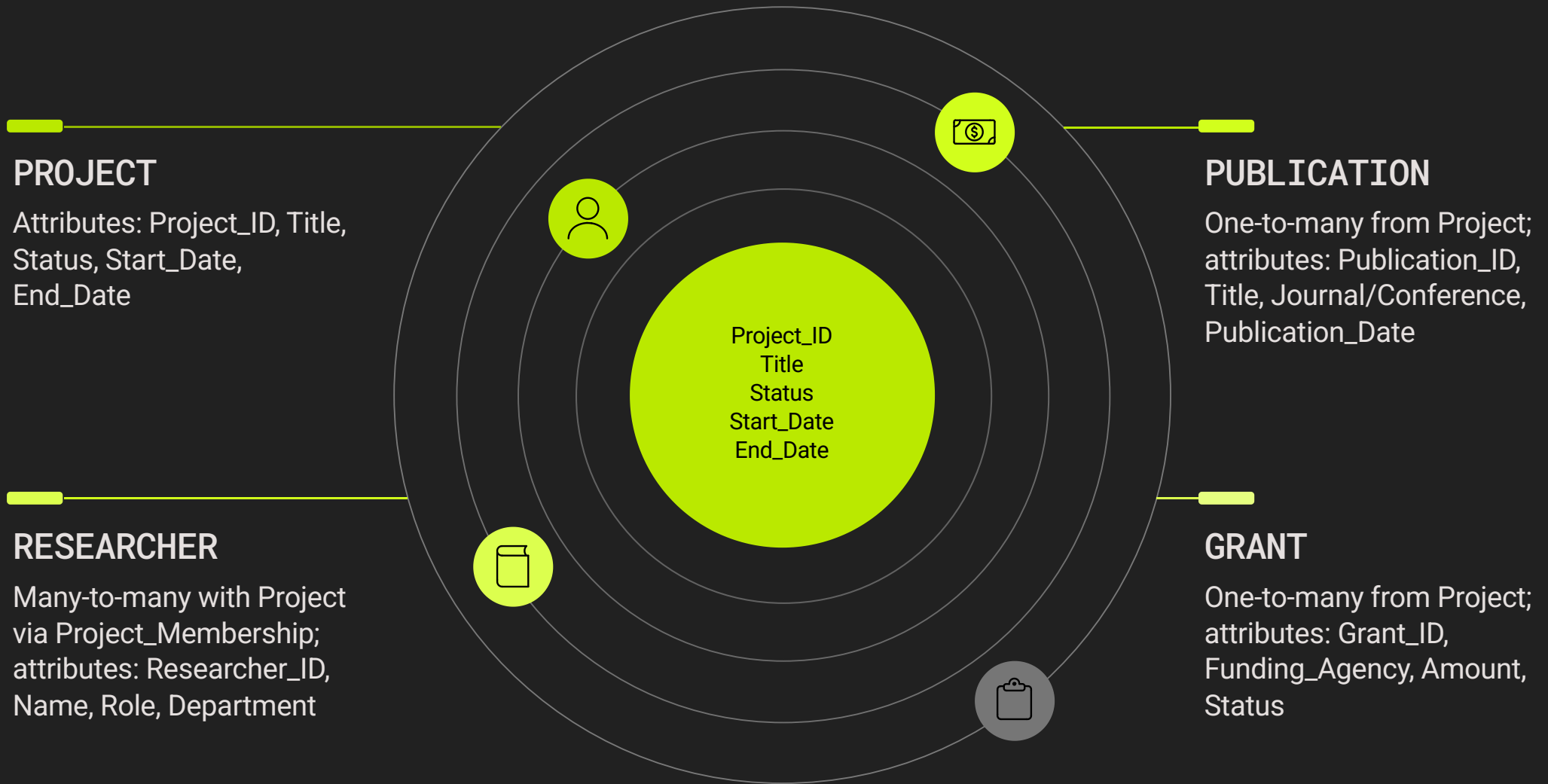
Duplication

Multiple versions of the same record lead to conflicting metadata and wasted effort reconciling entries.

Security & access

Ad-hoc file sharing exposes sensitive drafts and budget details; role-based access is absent.

Database & entity model



Core entities: Project, Researcher, Publication, Grant. The database uses primary keys, foreign keys and a junction table for project-researcher relationships.

Example research project – record snapshot

Field	Example
Project ID	PRJ-001
Title	AI-Based Predictive Maintenance
Lead Researcher	Dr. A. Kumar
Grant	2,50,000 Rs.
Status	Active

Problem formulation – Input, Output & Objective

Inputs

- Research project details
- Researcher details
- Publication details
- Grant details

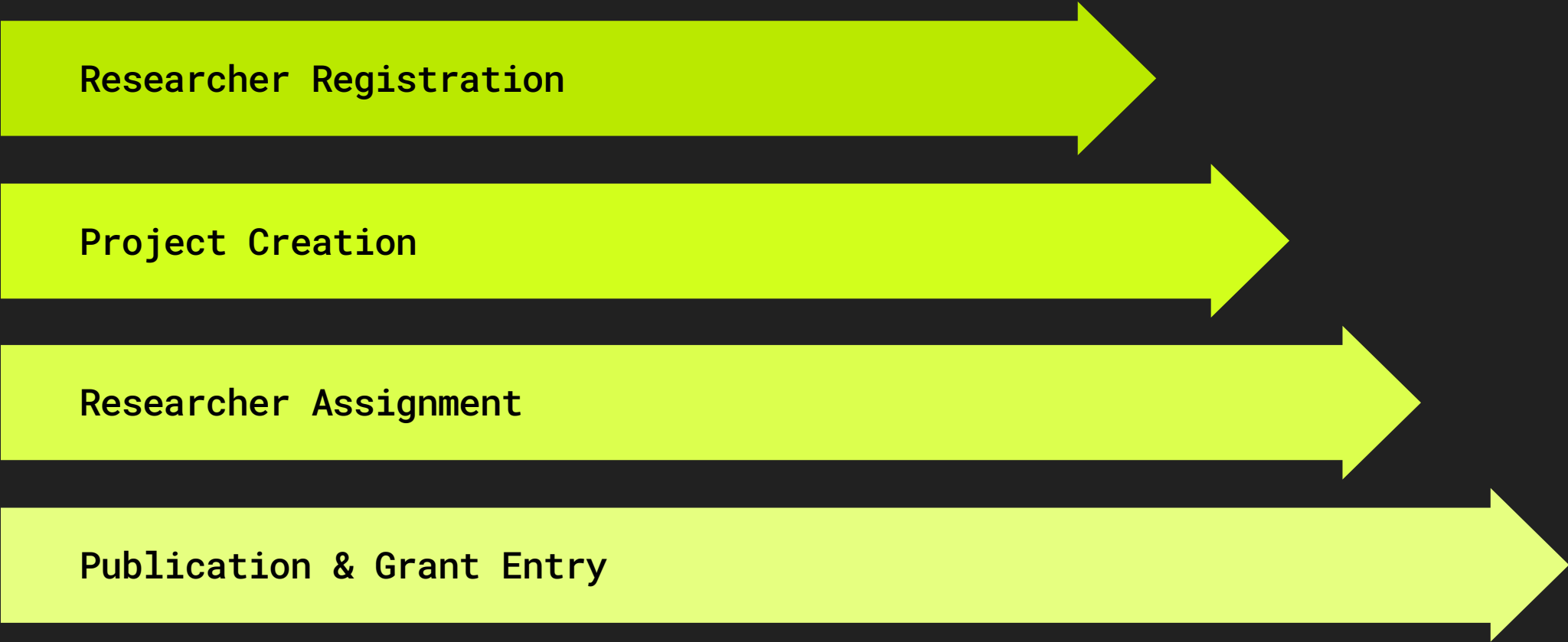
Outputs

- Project records
- Researcher-project associations
- Publication records
- Grant status and funding details
- Research management reports

Objective

Design a centralized relational database to store, organize and retrieve research project, publication and grant information efficiently.

Proposed DBMS solution & workflow



Researcher Registration

Project Creation

Researcher Assignment

Publication & Grant Entry

The DBMS stores research data in related tables and supports efficient data entry, retrieval, updates and reporting.

Alternative approaches – comparison

Approach	Pros	Cons
Spreadsheets	Easy to use	Duplication & poor relationships
Manual Records	Simple	Difficult to search/report
Relational DBMS	Structured, secure & scalable	Requires implementation



Why Relational DBMS?

- Maintains relationships between projects, researchers, publications and grants.
- Reduces duplicate and inconsistent data.
- Supports primary keys and foreign keys for data integrity.
- Allows efficient SQL-based search and reporting.

Why is it suitable?

A relational DBMS provides a structured and centralized way to manage research information and generate reports.

Database Design & Evaluation

Database Design

- Normalized relational tables
- Primary and foreign keys
- Referential integrity
- Reduced data duplication

Database Operations

- Project search
- Researcher-project search
- Publication retrieval
- Grant status retrieval

Evaluation Metrics

- Data accuracy
- Query response time
- Duplicate prevention
- Constraint validation
- Report accuracy

Test Cases

- Project creation
- Researcher assignment
- Publication linking
- Grant status update
- Invalid/duplicate data entry

Project roadmap – reviews & milestones

Review I – Problem & Design

- Problem statement
- Requirements
- ER diagram
- Relational schema

Review II – Implementation

- Database creation
- Tables & relationships
- SQL queries
- Sample data

Final Review – Complete Solution

- Testing
- Reports
- Results
- Demonstration
- Final documentation